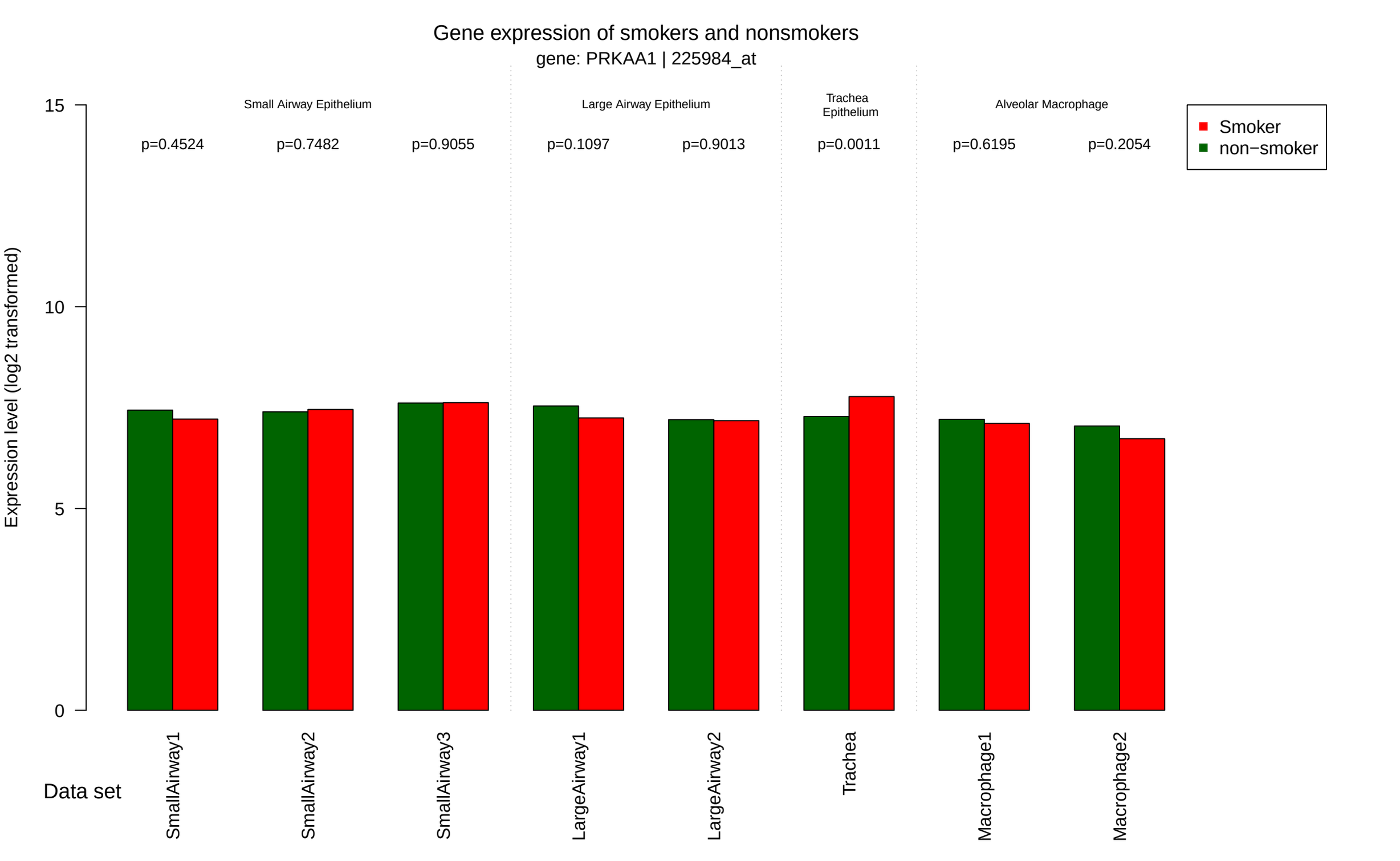
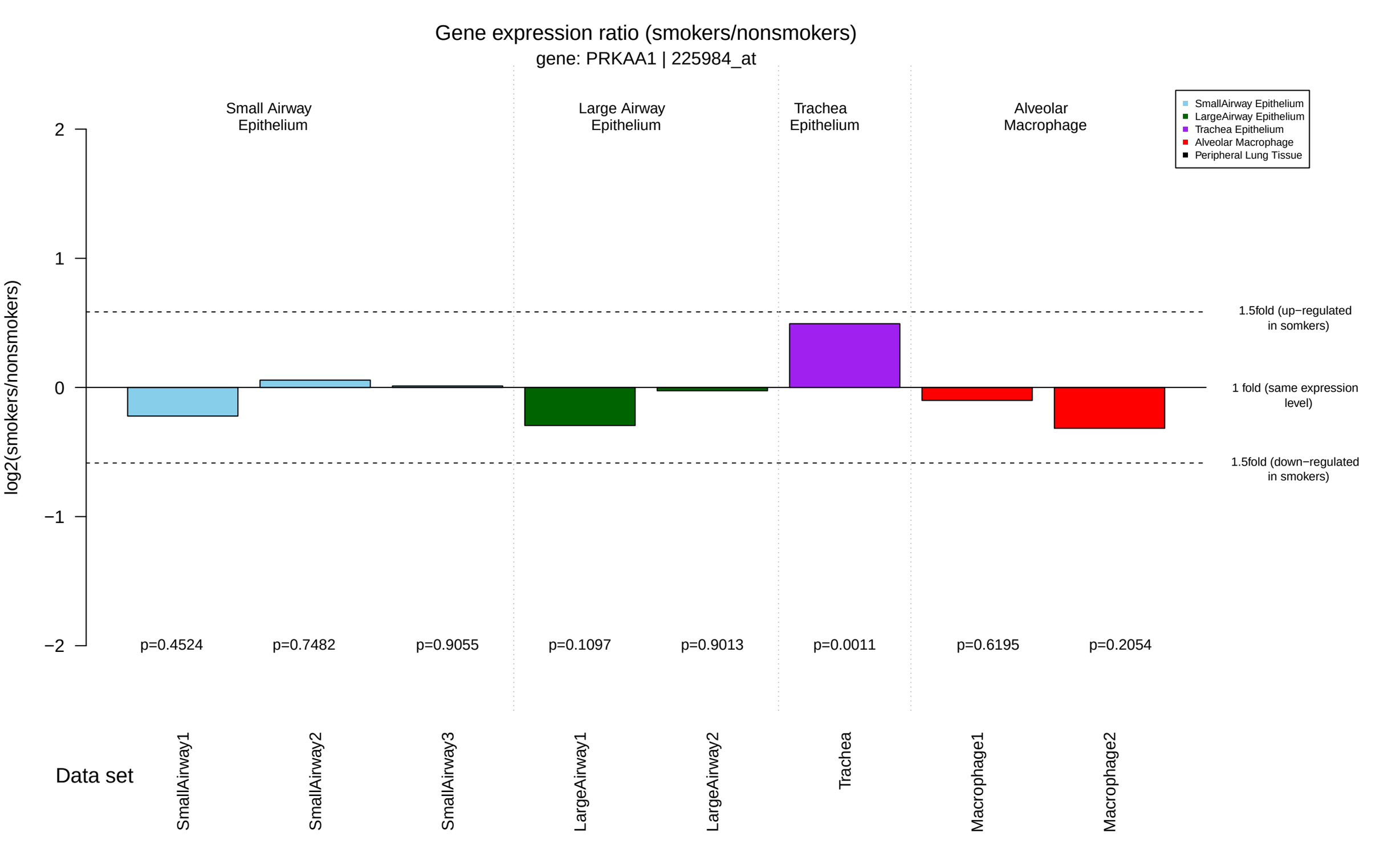
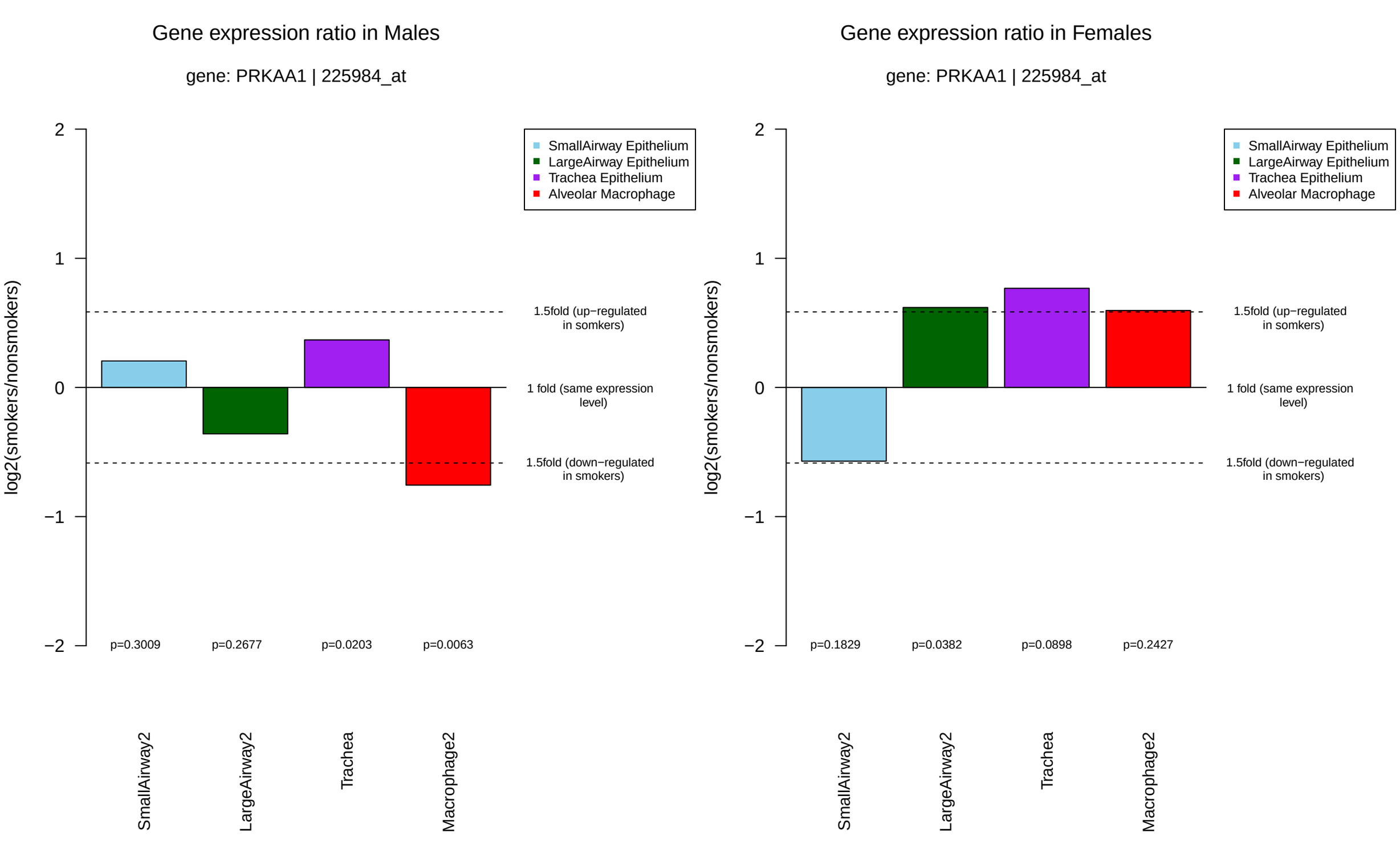
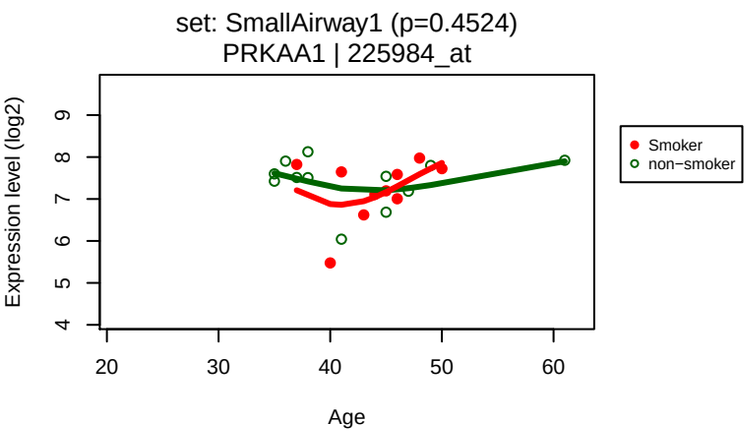
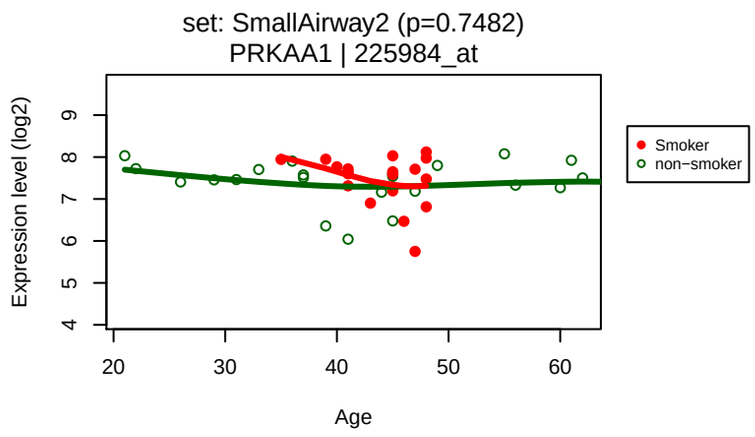
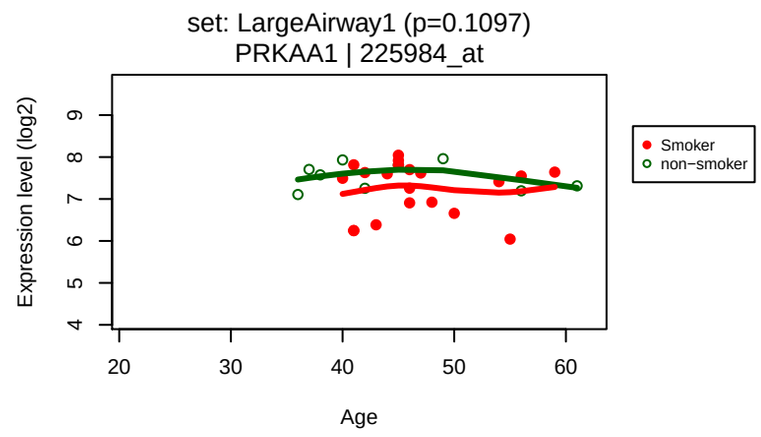

Gene expression of smokers and nonsmokers
gene: PRKAA1 | 225984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4524
p=0.7482
p=0.9055
p=0.1097
p=0.9013
p=0.0011
p=0.6195
p=0.2054
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

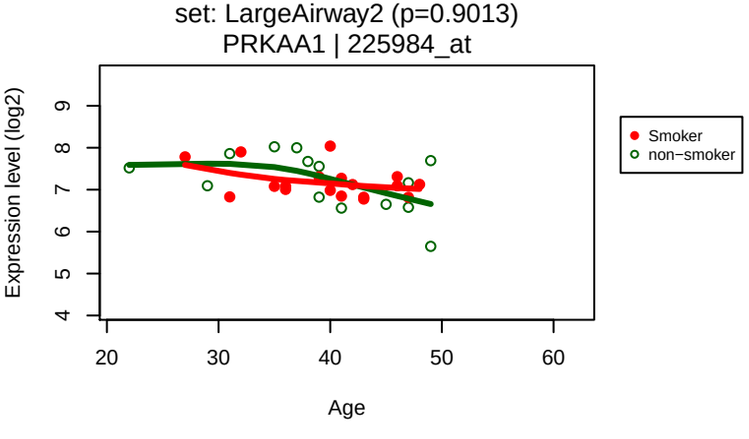

Gene expression ratio (smokers/nonsmokers)
gene: PRKAA1 | 225984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4524
p=0.7482
p=0.9055
p=0.1097
p=0.9013
p=0.0011
p=0.6195
p=0.2054
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

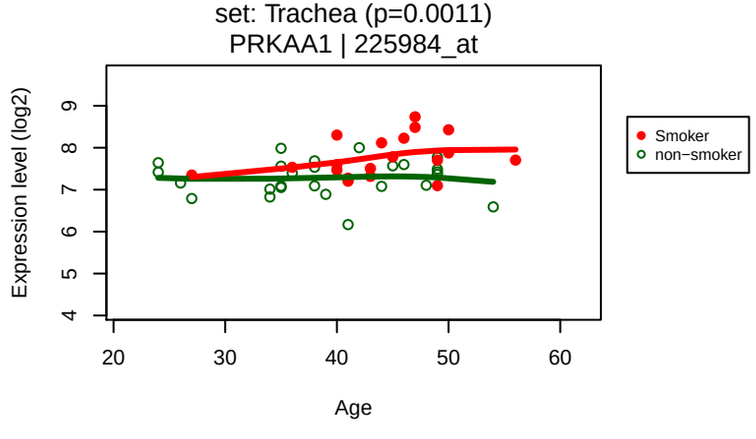

Gene expression ratio in Males
gene: PRKAA1 | 225984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3009
p=0.2677
p=0.0203
p=0.0063
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAA1 | 225984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1829
p=0.0382
p=0.0898
p=0.2427
SmallAirway2
LargeAirway2
Trachea
Macrophage2

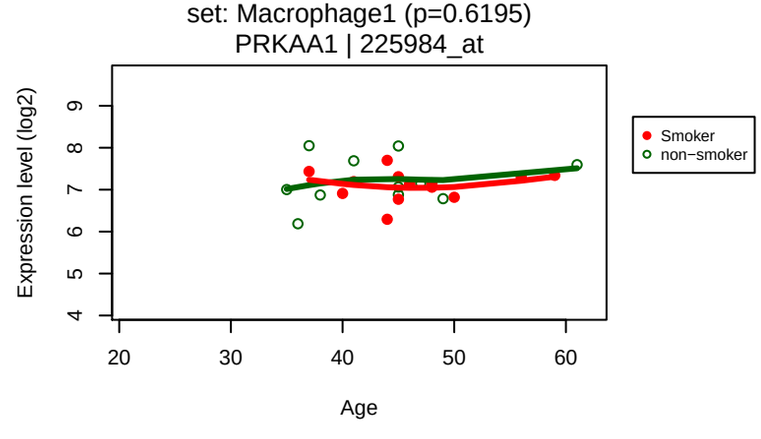

set: SmallAirway1 (p=0.4524)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker

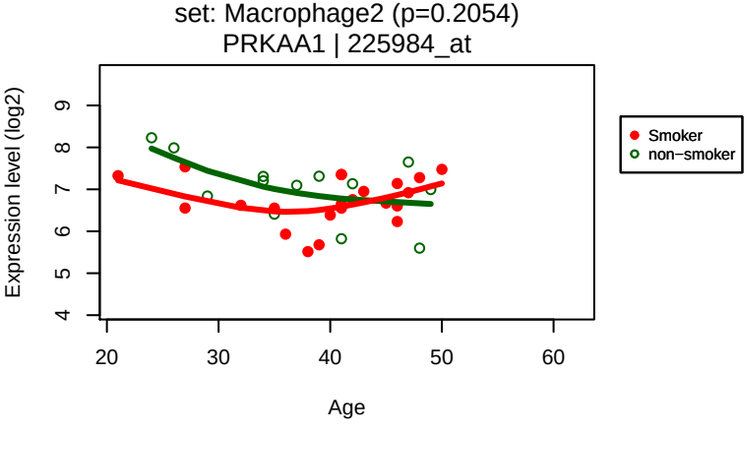

set: SmallAirway2 (p=0.7482)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker

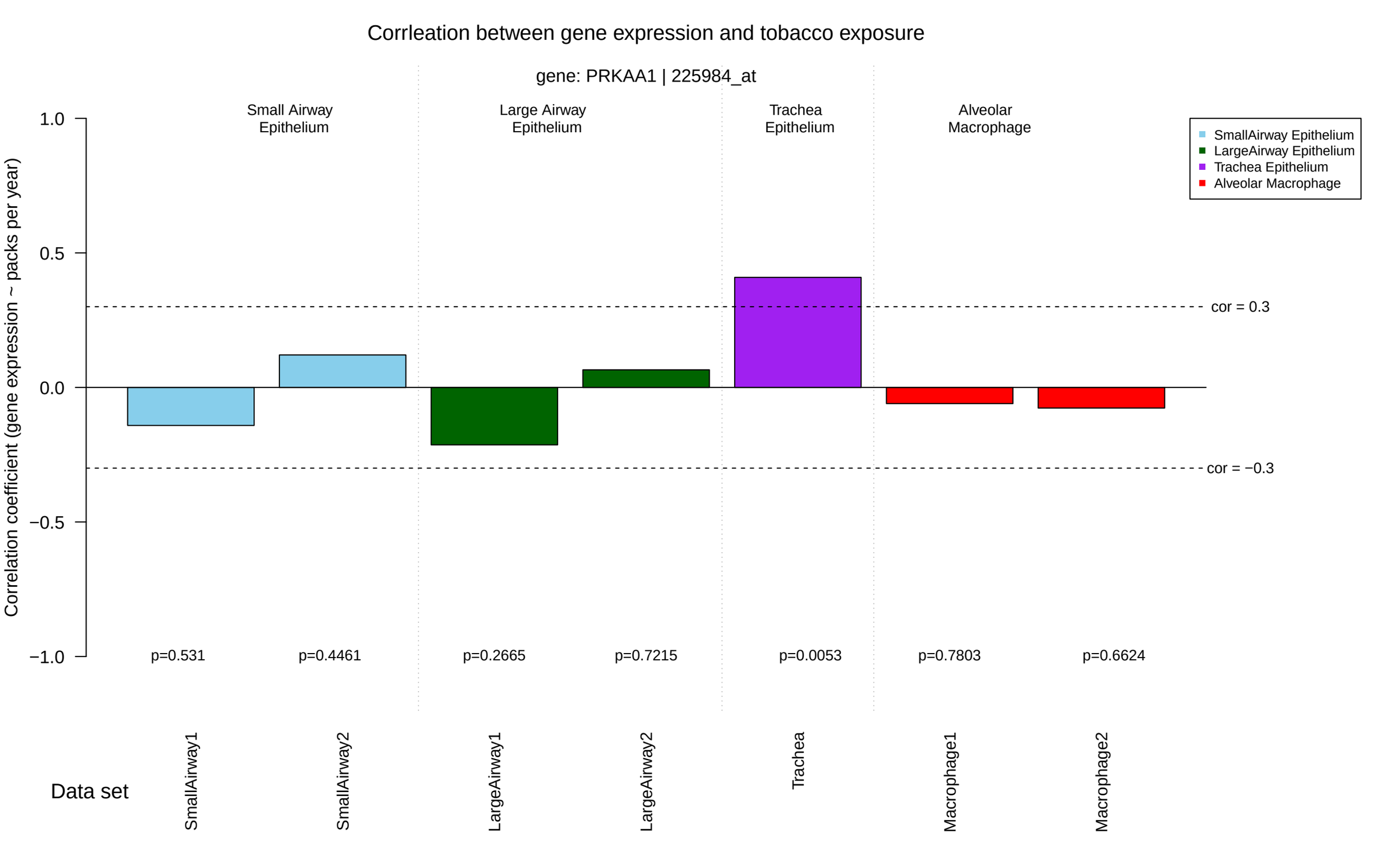

set: LargeAirway1 (p=0.1097)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9013)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6195)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2054)
PRKAA1 | 225984_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAA1 | 225984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.531
p=0.4461
p=0.2665
p=0.7215
p=0.0053
p=0.7803
p=0.6624
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2